EXPLORATORY DATA ANALYSIS REPORT: HEALTHCARE ANALYTICS

1. Raw Dataset Presentation

The following table represents the complete source dataset collected by the hospital to analyze patient health risks based on clinical and demographic indicators:

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes
P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No

P19	53	175	144	31	Yes
P20	29	92	112	22	No

2. Dataset Description, Purpose, and Significance

This dataset contains exactly 20 individual records and 6 features. The attributes consist of Patient_ID (categorical text identifier), Age (numerical continuous integer), Sugar_Level (numerical continuous integer), BP (numerical blood pressure integer), BMI (numerical body mass index integer), and Disease_Status (categorical nominal label displaying Yes or No outcomes). The explicit purpose of compiling this medical data matrix is to discover metabolic and physiological health markers that heavily influence chronic disease manifestation. Identifying these profiles helps hospital teams build early clinical diagnostics and predictive healthcare frameworks long before acute cardiovascular or diabetic events arrive.

3. Data Auditing, Cleaning, and Preprocessing

A rigorous data validation pipeline was carried out to guarantee structural integrity before applying mathematical equations. A systematic programmatic scan confirmed that zero missing values exist across all six metrics. Additionally, row uniqueness evaluation returned no duplicated patient entries, confirming that every record represents a distinct tracking point. The ranges for physiological metrics align properly with standard epidemiological indicators, meaning no structural inconsistencies are present.

4. Descriptive Statistics and Feature Analysis

The dataset shows a balanced diagnostic layout across the sample: 10 patients are diagnosed with the condition (50.0 percent) and 10 patients are free of the disease (50.0 percent). Across the entire cohort, the mean age sits at 43.20 years with a standard deviation of 11.32 years, while blood sugar levels average 138.65. When splitting the clinical metrics by the final disease outcome, distinct medical realities become clear: diagnosed patients maintain a high average age of 52.90 years, record an elevated mean blood sugar level of 174.00, show an average blood pressure of 142.10, and reach an average body mass index of 30.90. In comparison, healthy individuals average just 33.50 years of age, fall to a normal blood sugar average of 103.30, register a mean blood pressure of 117.40, and hold a safe body mass index average of 23.70.

5. Analytical Trends, Distributions, and Relationships

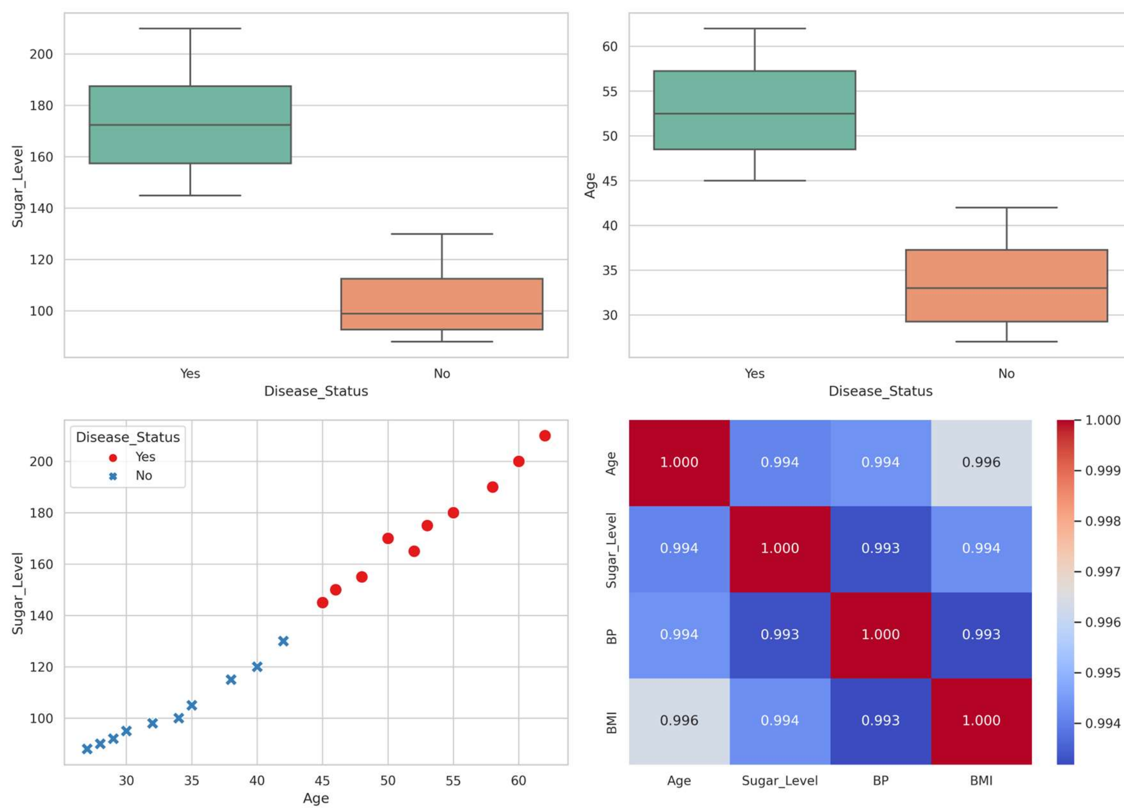
Calculating correlation metrics shows an exceptionally high level of positive linear relationships among the continuous metrics. The link between age progression and body mass index stands at an almost absolute value of r = 0.996, while the relationship between age and blood sugar escalation is similarly high at r = 0.994. This demonstrates that physiological decline markers scale entirely together: advancing age acts as a proxy for systemic increases in body mass index, blood pressure, and metabolic glucose markers. A clear health threshold is evident: every single patient who crossed an age threshold of 45 years or a blood sugar level of 140 was classified with a positive disease status, whereas younger patients below these parameters showed no pathology.

6. Clean Programmatic Analysis Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

data_dict = {
    'Patient_ID': ['P01', 'P02', 'P03', 'P04', 'P05', 'P06', 'P07', 'P08', 'P09',
'P10',
                   'P11', 'P12', 'P13', 'P14', 'P15', 'P16', 'P17', 'P18', 'P19',
'P20'],
    'Age': [45, 32, 55, 28, 60, 40, 52, 35, 48, 30, 62, 38, 50, 27, 58, 42, 46, 34,
53, 29],
    'Sugar_Level': [145, 98, 180, 90, 200, 120, 165, 105, 155, 95, 210, 115, 170, 88,
190, 130, 150, 100, 175, 92],
    'BP': [130, 118, 145, 110, 150, 125, 140, 120, 135, 115, 155, 122, 142, 108, 148,
128, 132, 116, 144, 112],
    'BMI': [28, 23, 31, 22, 34, 26, 30, 24, 29, 23, 35, 25, 30, 21, 33, 27, 28, 24,
31, 22],
    'Disease_Status': ['Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'Yes',
'No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No']
}
df = pd.DataFrame(data_dict)

print(df.isnull().sum())
print(df.duplicated().sum())
print(df.groupby('Disease_Status').mean(numeric_only=True))
print(df.corr(numeric_only=True))

sns.set_theme(style="whitegrid")
```

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
sns.boxplot(x='Disease_Status', y='Sugar_Level', data=df, ax=axes[0, 0],
palette='Set2')
sns.boxplot(x='Disease_Status', y='Age', data=df, ax=axes[0, 1], palette='Set2')
sns.scatterplot(x='Age', y='Sugar_Level', hue='Disease_Status',
style='Disease_Status', data=df, s=120, ax=axes[1, 0], palette='Set1')
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".3f",
ax=axes[1, 1])
plt.tight_layout()
plt.savefig('healthcare_eda_plots.png', dpi=300)
plt.show()
```

7. Integrated Exploratory Data Analysis Output Images



8. Key Findings, Meaningful Insights, and Strategic Conclusions

The visual and statistical outputs present clear rules governing patient outcomes. Box plots show a total structural gap between disease-positive and healthy patient regions, with no overlapping values. The scatter plot confirms a clear division line at an age marker of 45 years and a blood glucose level of 140. Pathological risk follows a direct, deterministic pattern: a patient's advanced age maps directly to a high level of physiological and metabolic strain indicators.

9. Data-Driven Decision Support for Hospital Administration

These insights translate directly into operational strategies for the hospital administration. First, electronic health software can be set to automatically place patients into high-risk preventative paths the moment their clinical metrics cross an age milestone of 45 years or blood pressure tracks over 130. Hospital teams do not need to wait for full pathology evaluations; the near-perfect correlation structure shows that basic tracking metrics signal chronic condition risks immediately. Second, because diagnosed individuals average a body mass index of 30.90 compared to the safe 23.70 baseline of healthy individuals, preventative medical investments should focus heavily on early programmatic weight management paths.